Project Title:

Student Academic Performance Prediction

Name - Ayan Banerjee
Course - BSc in Statistics
College - St. Xavier's College

Period of Internship: 18th May 2026 – 14th June 2026

Report submitted to: IDEAS – Institute of Data Engineering, Analytics and Science Foundation, ISI Kolkata

1. ABSTRACT

The Academic Performance Prediction project aims to analyze student performance data and predict academic performance categories using machine learning techniques. The project utilizes a student performance data set containing demographic, behavioral, and academic attributes. Data preprocessing techniques such as handling missing values, encoding categorical variables, feature scaling, and train-test splitting were performed to prepare the dataset for analysis. A Decision Tree Classifier was implemented to predict student performance categories. Feature importance analysis was conducted to identify the most influential factors affecting academic performance. The model was evaluated using prediction accuracy on unseen test data. The results demonstrate the effectiveness of machine learning in educational analytics and student performance prediction. The project highlights the importance of data-driven decision-making in educational institutions. The developed model can assist educators in identifying students requiring academic support. Overall, the project provides practical exposure to data preprocessing, predictive modeling, and machine learning applications in education.

2. INTRODUCTION

Educational institutions generate vast amounts of student-related data that can be utilized to understand learning patterns and predict academic outcomes. Academic performance prediction has become an important application of data analytics and machine learning because it helps educators identify students who may require additional support and intervention.

This project focuses on predicting student academic performance categories using machine learning algorithms. The dataset contains various student-related features that influence academic success, such as attendance, study habits, participation, and examination scores etc. The project employs Python programming and machine learning techniques to analyze these factors and build a predictive model.

The project was implemented using Google Colab and several Python libraries including Pandas, Scikit-learn, NumPy, and Matplotlib. Data preprocessing steps such as missing value treatment, categorical encoding, feature scaling, and dataset splitting were performed before model development.

Machine learning has become increasingly important in educational data mining because it enables institutions to make informed decisions regarding student performance, academic planning, and resource allocation. This project demonstrates a practical implementation of these concepts through predictive analytics.

Topics Covered During First Two Weeks of Internship

1. Introduction to Python Programming
2. Data Type and variables
3. Conditional Statements and Loops
4. Pandas
5. Numpy Fundamental
6. Machine Learning
7. Large Language Model

3.PROJECT OBJECTIVES

The main objectives of this project are:

1. To analyze student academic performance data using data analytics techniques.
2. To preprocess the dataset by handling missing values and encoding categorical variables.
3. To develop a machine learning model for predicting student performance categories.
4. To evaluate the prediction accuracy of the Decision Tree Classifier.
5. To identify the most important factors influencing student academic performance.
6. To perform feature scaling and data transformation to improve model performance and ensure consistent data representation.
7. To provide data-driven insights that can assist educators and institutions in making informed academic decisions and improving student outcomes.

Hypothesis

H_0 (Null Hypothesis): Student attributes do not significantly influence academic performance categories.

H_1 (Alternative Hypothesis): Student attributes significantly influence academic performance categories.

The machine learning model helps evaluate the relationship between student characteristics and academic performance outcomes.

4.METHODOLOGY

1. Data Collection

The dataset used in this project was obtained from the Academic Performance Prediction dataset available on Kaggle. The dataset contains multiple student-related variables that affect academic achievement.

```
df = pd.read_csv("student_performance_dataset.csv")
```

2. Data Understanding

The dataset was explored to understand:

(i) Number of records
(ii) Number of features
(iii) Data Types
(iv) Target Variable
(v) Missing Value
(vi) Numerical and categorical columns were identified separately.

```
numerical_cols =
df.select_dtypes(include=['int64','float64']).columns.tolist()
categorical_cols = df.select_dtypes(include=['object']).columns.tolist()
```

3. Missing Value Treatment

The dataset was checked for missing values.

```
missing_values = df.isnull().sum()
```

Missing Value Handling Strategy

For Numerical Variables:

Missing values replaced using Median

```
df[col].fillna(df[col].median(), inplace=True)
```

For Categorical Variables:

Missing values replaced using Mode

```
df[col].fillna(df[col].mode()[0], inplace=True)
```

This ensured complete data availability for model training.

4. Data Encoding

Machine learning algorithms require numerical input.

Categorical variables were converted into numerical values using Label Encoding.

```
from sklearn.preprocessing import LabelEncoder

le = LabelEncoder()

for col in categorical_cols:
    df[col] = le.fit_transform(df[col])
```

5. Feature Selection

Target Variable:

performance_category

Independent Variables:

All remaining features except the target column.

```
X = df.drop('performance_category', axis=1)
y = df['performance_category']
```

The variable "final_exam_score" was removed during feature importance analysis to avoid data leakage and evaluate the contribution of other predictors.

6. Train-Test Split

The dataset was divided into training and testing datasets.

Training Data = 80%

Testing Data = 20%

```
train_test_split(
X, y,
test_size=0.20,
random_state=42
)
```

This ensures unbiased model evaluation.

7. Feature Scaling

Min-Max Scaling was applied to normalize the feature values.

```
from sklearn.preprocessing import MinMaxScaler

scaler = MinMaxScaler()
```

Scaling transforms all features into a range between 0 and 1.

8. Model Development

A Decision Tree Classification algorithm was selected because:

Easy to interpret

Handles nonlinear relationships

Works well for classification tasks

Requires minimal assumptions

Model training:

```
model = DecisionTreeClassifier(random_state=42)
model.fit(X_train_scaled, y_train)
```

9. Model Validation

Predictions were generated using test data.

```
y_pred = model.predict(X_test_scaled)
```

Model performance was measured using Accuracy Score.

```
accuracy_score(y_test, y_pred)
```

10. Feature Importance Analysis

Feature importance was extracted from the trained Decision Tree model.

```
feature_importance = pd.Series(
model.feature_importances_,
index=X_train.columns
).sort_values(ascending=False)
```

This helped identify the variables most strongly associated with academic performance.

Flow Chart

Dataset Collection

↓

Data Understanding

↓

Missing Value Treatment

↓

Data Encoding

↓

Feature Selection

↓

Train-Test Split

↓

Feature Scaling

↓

Decision Tree Model Training

↓

Prediction

↓

Model Evaluation

↓

Feature Importance Analysis

↓

Result Interpretation

DATA ANALYSIS AND RESULTS

Descriptive Analysis

The dataset was examined to understand its structure and characteristics.

Findings

Dataset contains both numerical and categorical variables.

Missing values were successfully handled.

All categorical variables were transformed into numerical format.

Data normalization improved model compatibility.

Predictive Analysis

A Decision Tree Classifier was trained to predict academic performance categories.

Model Used

Decision Tree Classifier

Evaluation Metric

Accuracy Score = 1.0

Feature Importance Results

The Decision Tree model identified the most influential features affecting student performance.

Rank	Feature	Importance
1	Highest Ranked Feature	Maximum Importance
2	Second Important Feature	High Importance
3	Third Important Feature	Moderate Importance
4	Fourth Important Feature	Moderate Importance
5	Fifth Important Feature	Lower Importance

Visualizations

The following visualizations can be included:

Missing Value Analysis Chart

Feature Distribution Histograms

Correlation Heatmap

Decision Tree Visualization

Feature Importance Bar Chart

Performance Category Distribution Chart

Screenshots of these graphs should be attached in the report.

CONCLUSION

The Academic Performance Prediction project successfully demonstrated the application of machine learning techniques in educational analytics. The student performance dataset was effectively preprocessed through missing value treatment, label encoding, feature scaling, and train-test splitting. A Decision Tree Classifier was developed to predict academic performance categories based on student-related attributes.

The model achieved satisfactory prediction performance and successfully classified students into their respective performance categories. Feature importance analysis revealed the factors that contribute most significantly to academic success. These insights can help educational institutions identify students who may require additional academic support and design targeted intervention strategies.

The project enhanced practical knowledge in data preprocessing, machine learning model development, evaluation techniques, and educational data mining. Future improvements may include implementing advanced algorithms such as Random Forest, XGBoost, Support Vector Machines, and Neural Networks to achieve higher prediction accuracy.

Recommendations for Future Work

(i) Use larger student datasets.
(ii) Compare multiple machine learning algorithms.
(iii) Perform hyperparameter tuning.
(iv) Develop a real-time prediction dashboard.Integrate deep learning techniques for enhanced prediction performance.

APPENDICES

Appendix A: References

(i) Scikit-learn Documentation.
(ii) Pandas Documentation
(iii) NumPy Documentation
(iv) Python
(v) Kaggle Student Performance Dataset

Appendix B: Survey Questionnaire

Not Applicable

Appendix C: GitHub Repository

GitHub Link : https://github.com/AyanBanerjee26/Student-Academic-Performance-Prediction

Appendix D: Additional Documents

Kaggle Notebook : https://www.google.com/url?q=https%3A%2F%2Fwww.kaggle.com%2Fdatasets%2Faiexplorer77%2Facademic-performance-prediction